

S-DES: The Ultimate Colored Guide

Fully Visualized Steps - Split Pages for Clarity

1. Reference Tables

P10										P8							
1	2	3	4	5	6	7	8	9	10	1	2	3	4	5	6	7	8
3	5	2	7	4	10	1	9	8	6	6	3	7	4	8	5	10	9

IP								E/P							
1	2	3	4	5	6	7	8	1	2	3	4	5	6	7	8
2	6	3	1	4	8	5	7	4	1	2	3	2	3	4	1

P4				IP ⁻¹							
1	2	3	4	1	2	3	4	5	6	7	8
2	4	3	1	4	1	3	5	7	2	8	6

S-Boxes

S₀ :

1032

3210

0213

3132

S₁ :

0123

2013

3010

2103

2. Key Generation

Raw Key: 1010000010

Step 1	P10:	In	1010000010								
		Out	1000001100								

Step 2	LS-1: L:	In	10000	R:	In	01100					
		Out	00001		Out	11000					

Step 3	K1 (P8):	In	0000111000								
		Out	10100100								

Step 4	LS-2: L:	In	00001	R:	In	11000					
		Out	00100		Out	00011					

Step 5	K2 (P8):	In	0010000011								
		Out	01000011								

3. Encryption (Round 1)

Plaintext: 01110010

#	Stage	Details	
1	IP	In	01110010
		Out	10101001
$L_0 = 1010, R_0 = 1001$			
ROUND 1 (Key: 10100100)			
2	Exp & XOR	A. Exp:	In: 1001 (R0) Out: 11000011
		B. XOR:	In: 11000011 Ky: 10100100 =: 01100111
3	S-Boxes	S_0 :	In: 0110 Row/Col: R:00(0), C:11(3) Out: 10
		S_1 :	In: 0111 Row/Col: R:01(1), C:11(3) Out: 11
Combined: 1011			
4	P4 & XOR	A. P4:	In: 1011 Out: 0111
		B. XOR:	In: 1010 (L0) Ky: 0111 (P4) ← New R =: 1101
SWAP: Left ← 1001, Right ← 1101			

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3. Encryption (Round 2 & Final)

Current State: $L = 1001, R = 1101$

#	Stage	Details	
ROUND 2 (Key: 01000011)			
5	Exp & XOR	A. Exp:	In1101 (R)
			Out11101011
		B. XOR:	In11101011
			Ky01000011
=10101000			
6	S-Boxes	S0:	In1010
			Row/ColR:10(2), C:01(1)
			Out10
		S1:	In1000
Row/ColR:10(2), C:00(0)			
Out11			
Combined: 1011			
7	P4 & XOR	A. P4:	In1011
			Out0111
		B. XOR:	In1001 (L)
			Ky0111 (P4) ← Final L
=1110			
8	Final	Combine L2R2: 11101101 IP Inverse: 01110111	

4. Decryption (Round 1)

Ciphertext: 01110111

#	Stage	Details							
1	IP	In01110111							
		Out11101101							
		$L = 1110, R = 1101$							
ROUND 1 (Key: 01000011)									
2	Exp & XOR	A. Exp:	In1101 (R)						
			Out11101011						
		B. XOR:	In11101011						
			Ky01000011						
			=10101000						
3	S-Boxes	S_0 :	In1010			S_1 :	In1000		
			Row/ColR:10(2), C:01(1)				Row/ColR:10(2), C:00(0)		
			Out10				Out11		
		Combined: 1011							
		4	P4 & XOR	A. P4:	In1011				
Out0111									
B. XOR:	In1110 (L)					$\leftarrow$ New R			
	Ky0111 (P4)								
	=1001								
SWAP: Left $\leftarrow$ 1101, Right $\leftarrow$ 1001									

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4. Decryption (Round 2 & Final)

Current State: $L = 1101, R = 1001$

#	Stage	Details			
ROUND 2 (Key: 10100100)					
5	Exp & XOR	A. Exp:	In	1001 (R)	
			Out	11000011	
		B. XOR:	In	11000011	
			Ky	10100100	
=	01100111				
6	S-Boxes	S ₀ :	In	0110	
			Row/Col	R:00(0), C:11(3)	
			Out	10	
		S ₁ :	In	0111	
Row/Col	R:01(1), C:11(3)				
Out	11				
		Combined: 1011			
7	P4 & XOR	A. P4:	In	1011	
			Out	0111	
		B. XOR:	In	1101 (L)	
			Ky	0111 (P4)	
=	1010				
		← Final L			
8	Final	Combine LR: 10101001 IP Inverse: 01110010			